THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM ONE ANNUAL EXAMINATION
032 CHEMISTRY
MARKING SCHEME NOVEMBER 2025

SECTION A (15 MARKS)

1. **(10 marks)**

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
C	B	B	B	C	B	B	D	B	C

2. **(05 marks)**

(i)	(ii)	(iii)	(iv)	(v)
B	D	A	E	F

SECTION B (70 MARKS)

3. (a) Difference between an element and a compound: **(04 marks)**
   - **Element:** A pure substance that cannot be broken down into simpler substances by chemical means. It consists of only one type of atom.
   - **Compound:** A substance formed when two or more different elements are chemically combined in a fixed ratio. It can be broken down into simpler substances by chemical means.

   (b) Two common laboratory hazards and their prevention:
   (i) **Hazard 1: Chemical spills.** Prevention: Always wear appropriate personal protective equipment (PPE) like gloves and safety goggles. Handle chemicals carefully, and know the location of spill kits and emergency showers. **(03 marks)**

   (ii) **Hazard 2: Broken glassware.** Prevention: Inspect glassware for cracks before use. Dispose of broken glassware in designated containers, not regular trash bins. Use caution when handling glassware. **(03 marks)**

4. (a) (i) It is applied to manufacture substances like paints, varnishes, cement, plastics and iron sheets. **(1½ Marks)**
   (ii) It is applied to manufacture fertilizer, pesticides and animal vaccines. **(1½ Marks)**
   (iii) It is applied to manufacture drugs, vaccines and food supplements. **(1½ Marks)**
   (iv) It is applied to manufacture products like fuels, lubricant (oil), coolants and tyres. **(1½ Marks)**

   (b) (i) Doctor
   (ii) Nurse
   (iii) Teacher
   (iv) Chemical engineer

(v) Laboratory technician **(Any 4 points, 1@ = 4 Marks)**

5. (a) Chromatography: Chromatography is a laboratory technique for separating a mixture. Its principle is based on the differential distribution of components between a stationary phase and a mobile phase. Components that are more soluble in the mobile phase or have less affinity for the stationary phase move faster, leading to separation. **(04 marks)**
   (b) Suitable methods for separating the following mixtures:
   (i) **Sand and salt:** Dissolve the salt in water, then filter out the sand. Evaporate the water to recover the salt. So, **dissolving, filtration, and evaporation**. **(02 marks)**
   (ii) **Water and kerosene:** Kerosene is immiscible with water and less dense. A **separating funnel** can be used to separate them. **(02 marks)**
   (iii)**Alcohol and water:** Alcohol and water are miscible liquids with different boiling points. **Fractional distillation** can be used to separate them. **(02 marks)**

6. (a) (i) Periodicity is the repetition of similar chemical properties of elements after a certain specific interval in a given arrangement. **(02 Marks)**
   (ii) Periodic table is a tabular display of the chemical elements, which are arranged by atomic number, electron configuration, and recurring chemical properties **(02 Marks)**

(b)
   (i) Alkali metals are Group 1 elements; they are very reactive (have one electron in the outer valence shell). Example; Lithium, sodium and potassium.
   (ii) Alkaline earth metals are group 2 elements; they are very reactive (have two electrons in the outer valence shell). Example; Beryllium, magnesium and calcium.
   (iii)The noble gases are group 8 elements. They have full outer shells. They have stable duplet (2 electrons) or octet (8 electrons) in their outermost shells which makes them very stable and therefore, unreactive. Examples are Helium and Neon **(02@ =06 Marks)**

7. (a).
   (i) Class B fires**:** These fires involve flammable liquids such as petrol, kerosene, oil, alcohol, ether, vanishes, etc. For small fires, a fire blanket or sand may be used. If the fire is large, use foam, dry powder or carbon dioxide extinguisher. Water should not be used on class B fires because the burning material, being lighter than water, will just float and spread the fire further.
   (ii) Class C fires: The burning material involves flammable gases such as hydrogen, acetylene, coal gas, butane, methane, and propane. The best fire extinguishers to use in fighting against these fires are foam, dry powder or carbon dioxide extinguishers. It is important to turn off the gas supply, and spray water on the gas tank to cool it down. **(@2= 04 marks)**

   (b). A fire will continue or start to burn if these components are present:
   (i) **Fuel:** This refers to any combustible material be it solid, liquid or gaseous material provided it can catch fire and burn. You can stop fire by removing the combustible material from the path of fire.
   (ii) **Oxygen (air):** Oxygen supports combustion. A fuel will only burn if there is sufficient supply of oxygen. You can extinguish fire by displacing, or taking away oxygen supply from the fire or by blocking the gas supply to the fire.
   (iii) **Heat:** For a fire to start burning, the temperature should be at the kindling point of that fuel or above it. Every fuel has its own kindling or ignition point. Below the

ignition point, the fuel will not catch fire. You can put out fire by lowering the temperature below the kindling point of a particular fuel. **(@2 = 06 marks)**

8. Laboratory rules:
   (i) Do not enter the laboratory without permission from your teacher or laboratory technician.
   (ii) Wear safety goggles all the time while in the laboratory.
   (iii) Never run unnecessarily in the laboratory. **Any ten 01@ = 10 marks**
   (iv) Eating, drinking, and smoking are strictly prohibited in the laboratory.
   (v) Don't perform any experiment not authorized by your teacher or lab technician.
   (vi) Never taste anything.
   (vii) Notify your teacher or technician immediately in case of an accident
   (viii) Know what chemicals you are using, carefully read the label twice before taking anything from the reagent bottle. Do not interchange labels.
   (ix) Pour more concentrated solutions into less concentrated solutions to avoid violent reactions. For example, always add acid to water; not water to acid.
   (x) If chemicals accidentally splash onto your skin or eyes, flush immediately with plentiful amounts of water and report to your teacher or lab technician.
   (xi) Never point a test tube or vessel that you are heating at yourself or your colleague.
   (xii) Make sure all the burners are turned off before leaving the laboratory.
   (xiii) Never use broken glassware. If glassware breaks, inform your teacher and dispose of glassware in the litter bin.

9. (a) Solution is a uniform mixture (homogeneous mixture) of two or more substances. **01 mark**
   Examples: salt in water solution, Alcohol in water solution. **0½@= 1 mark**

(b)

	Mixture	**Compound**
1	The components of a mixture can be separated by physical means, example, by filtering.	The components of a compound can be separated by chemical means only.
2	The composition of a mixture can vary widely, example, a mixture of 20g of sand with 1g of salt or vice versa.	Compounds are fixed in their compositions by mass of elements present, example, there are always 2 atoms of hydrogen to 1 atom of oxygen in a molecule of water
3	Formation of mixture is not usually accompanied by heat changes.	Formation of a compound is usually accompanied by heat changes.
4	Properties of a mixture are the sum of the properties of the individual constituents of the mixture.	The properties of a compound are entirely different from those of its constituent elements. For example, water is a liquid whereas its constituent elements, hydrogen and oxygen, are both gases.

(Any four 02@ = 08 marks)

SECTION C (15 MARKS)

10. The Importance of Changing One State of Matter to Another
    (i) Separation of mixtures: It enables separation of mixtures can through such processes as distillation, sublimation, evaporation and condensation.
    (ii) Industrial manufacture of products: Industrially, the process of distillation is applied in the production of pure substances such as beer and other alcoholic drinks such as wine. The manufacturing process involves boiling, evaporating and condensation.
    (iii) Refining of petroleum (crude oil): Crude oil contains organic liquid components, each with a different boiling point. In the refinery, the components with lower boiling points evaporate first and get separated out, leaving those with higher boiling points behind. In this way, we get various types of oil components (fractions) such as petrol, diesel, kerosene, lubricating oil, etc.
    (iv) Drying of crops and clothes: When you suspend your clothing on a cloth line to dry, the moisture in it is lost through evaporation. Likewise, farmers in the village often spread crops on the ground to dry.
    (v) Cooling of our bodies in hot weather: Cooling of our bodies in hot weather depends in the evaporation of sweat or water.
    (vi) Ice formation in refrigerators: You can freeze water into ice and then use the resulting ice for cooling the beverage or preserving foods.
    (vii) Melting metals to make alloys: In metallurgical industries, need may arise to mix two or more metals (alloys) together. This is only possible, where two or more metals are first melted at high temperatures into liquids. Then the resulting liquid metals are mixed in appropriate proportions. This is followed by cooling down the mixture to a solid alloy.
    (viii) Formation of rain: Rain is mainly formed through the process of evaporation and condensation. Water vapour, evaporating mostly from water bodies (oceans, seas, lakes, rivers, ponds, etc), land and plants rises up to the sky. As it rises, it cools down and condenses into tiny droplets

Introduction=02mark
Main body 02 marks Total= 12 Marks
Conclusion=01 marks